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Assignment 3 SDG-Based Questions – CO3

Course Code ITA06
Course Name Machine Learning

concept to a real-world application aligned with the United Nations Sustainable Development Goals (SDGs).

Design a Bayes classifier to predict machine failures in manufacturing industries. for these give code to run in googlee collab i wantt graph represent also

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.datasets import make_classification
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import GaussianNB
from sklearn.metrics import (
    accuracy_score,
    confusion_matrix,
    classification_report,
    roc_curve,
    auc
)
```

```
# Synthetic Manufacturing Dataset
```

```
X, y = make_classification(
    n_samples=1000,
    n_features=6,
    n_informative=5,
    n_redundant=0,
    random_state=42
)
```

```
columns = [
    "Temperature",
    "Pressure",
    "Vibration",
    "Humidity",
    "Power_Consumption",
    "Machine_Age"
]
```

```
df = pd.DataFrame(X, columns=columns)
df["Machine_Failure"] = y
```

```
print(df.head())
```

	Temperature	Pressure	Vibration	Humidity	Power_Consumption	Machine_Age	\
0	3.329341	0.387315	1.616660	-0.223832	-1.563391	0.319790	
1	1.068028	1.051222	1.867333	0.566350	-0.370503	0.582681	
2	0.030641	1.652172	0.380666	-1.201240	1.276612	0.804256	
3	1.297464	0.157081	-0.934564	0.120242	-0.858308	1.099487	
4	0.310941	3.862320	-0.692832	2.753747	-0.154222	-0.842906	

```
Machine_Failure
0      1
1      0
2      1
3      1
4      0
```

```
X = df.drop("Machine_Failure", axis=1)
y = df["Machine_Failure"]
```

```
X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.25,
    random_state=42
)
```

```
model = GaussianNB()

model.fit(X_train, y_train)

prediction = model.predict(X_test)
```

```
accuracy = accuracy_score(y_test, prediction)

print("Accuracy :", accuracy)
```

```
print("\nClassification Report\n")
print(classification_report(y_test, prediction))
```

Accuracy : 0.796

Classification Report

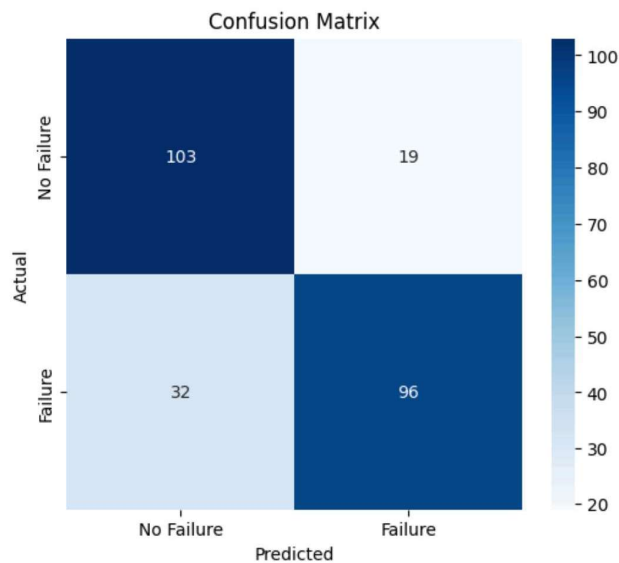
	precision	recall	f1-score	support
0	0.76	0.84	0.80	122
1	0.83	0.75	0.79	128
accuracy			0.80	250
macro avg	0.80	0.80	0.80	250
weighted avg	0.80	0.80	0.80	250

```
cm = confusion_matrix(y_test, prediction)

plt.figure(figsize=(6,5))

sns.heatmap(
    cm,
    annot=True,
    fmt="d",
    cmap="Blues",
    xticklabels=["No Failure","Failure"],
    yticklabels=["No Failure","Failure"]
)

plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.title("Confusion Matrix")
plt.show()
```



```
plt.figure(figsize=(6,5))

sns.countplot(
    x=df["Machine_Failure"],
    palette="Set2"
)

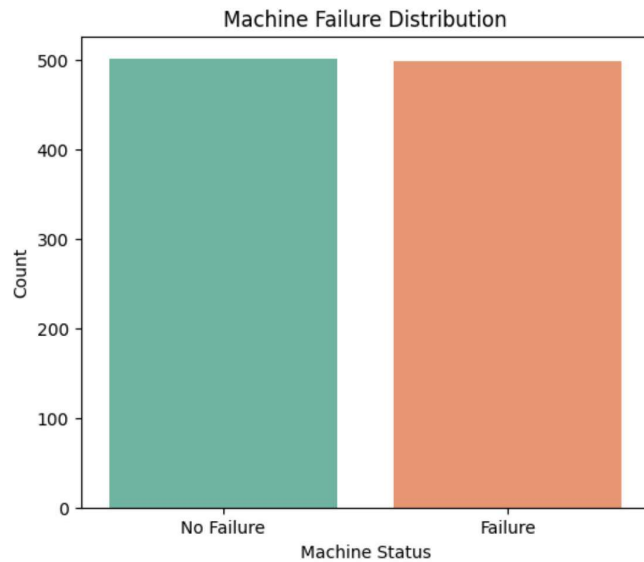
plt.xticks([0,1],["No Failure","Failure"])
plt.title("Machine Failure Distribution")
plt.xlabel("Machine Status")
plt.ylabel("Count")

plt.show()
```

```
/tmp/ipykernel_813/1499437806.py:3: FutureWarning:
```

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

```
sns.countplot(
```



```
prob = model.predict_proba(X_test)[:,-1]

fpr, tpr, threshold = roc_curve(y_test, prob)

roc_auc = auc(fpr, tpr)

plt.figure(figsize=(6,6))

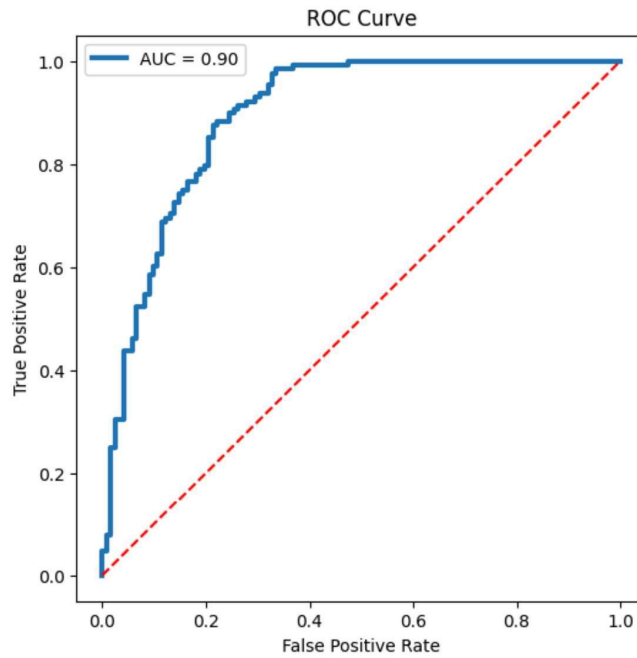
plt.plot(fpr, tpr,
         label="AUC = %.2f"%roc_auc,
         linewidth=3)

plt.plot([0,1],[0,1], 'r--')

plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve")

plt.legend()

plt.show()
```



```
new_machine = [[
    0.8,
    -0.4,
    1.2,
    0.5,
    -0.7,
    0.3
]]

result = model.predict(new_machine)

if result[0] == 1:
    print("Prediction : Machine Failure")
else:
    print("Prediction : Machine is Healthy")
```